

Tran Minh Duong

Junior AI Engineer

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About Me

I design, train, and deploy ML systems with a focus on reliability, monitoring, and real-world constraints. Experienced in RAG pipelines, LLM fine-tuning, model evaluation, containerized inference and cloud deployment.

Looking for AI Engineering / MLOps roles where I can ship and improve model delivery.

Education

- BSc** **La Rochelle University**, Informatique La Rochelle, France
• Thesis: "Historical Fidelity in OCR Post-Correction: A Comparative Study of Parameters Efficient Fine-Tuned LLMs and Seq2Seq Models"
• GPA: 17.05/20 ($\approx 3.89/4$) Sept 2025 – present
- BSc** **University of Science and Technology of Hanoi**, ICT (Double Degree) Hanoi, Vietnam
• GPA: 18.48/20 ($\approx 3.89/4$), ranked **1st in double degree cohort** [↗](#)
• Awarded 100% Merit Scholarship for 2 consecutive years Sept 2023 – July 2025

Skills

Programming Languages: Python (proficient), Java, C++, Bash, SQL

Framework / Libraries: PyTorch, Hugging Face transformers, Langchain, FastAPI, PyTest, Weights & Biases

Databases: PostgreSQL, Supabase, MongoDB, Neo4j (Graph), Pinecone (Vector), ChromaDB

Tools & DevOps: AWS, Docker, Terraform, Git, GitHub Actions, Prometheus, Grafana

Foreign Languages: English C1 (IELTS 7.5), French B2 (DELF B2)

Experience

- FPT Software Innovation**, Junior AI Engineer Hanoi, Vietnam
Design and deliver scalable AI-powered solutions for enterprise clients, focusing on LLM integrations and MLOps practices. June 2026 – present
- Develop and deploy custom Retrieval-Augmented Generation (RAG) and NLP pipelines to automate document processing and enhance enterprise search capabilities
 - Collaborate with cross-functional teams to integrate machine learning models into production cloud environments, ensuring high reliability and low latency
 - Optimize model inference and perform comprehensive evaluations using modern MLOps tools to prevent regression in production deployments
- L3i - La Rochelle Université**, AI Researcher La Rochelle, France
Conducted research on OCR Post-Correction for historical documents using modern LLMs. Mar 2026 – June 2026
- Researched modern LLM applications for historical document OCR post-correction
 - Optimized Qwen3 and BART via PEFT/QLoRA, decreasing OCR error rates by 50%
 - Analyzed metadata-aware vs. metadata-free prompt performance and assisted in writing the methodology for an academic conference paper
- ICTLab - USTH**, AI Research Assistant Hanoi, Vietnam
Contributed to research on **context-aware fire detection** using Visual Language Models Sept 2024 – Aug 2025
- Published the [fire context awareness dataset](#) [↗](#) to benchmark multimodal VLMs
 - Evaluated Qwen2.5VL, InternVL3 and other state-of-the-art models by implementing pipelines to compare various metrics and inference latency
 - Fine-tuned VLMs using **PEFT/LoRA** to boost models accuracy from 60% to 85%–90%, using different quantization configurations, tracked via **Weights & Biases**

Projects

RAG on AWS [↗](#)

End-to-End serverless hybrid RAG (Vector + Knowledge Graph) on AWS, entirely provisioned via Terraform

- Architected event-driven ingestion pipeline (Lambda/S3) achieving minimal cost (<\$1/mo)
- Automated tests in CI/CD using DeepEval (LLM-as-a-Judge) to prevent accuracy regression
- **Techstack:** Terraform, Amazon Cognito, Lambda, S3, ECR, Pinecone, Neo4j, DeepEval, Docker

Legal Contract Analyzer [↗](#)

RAG Chatbot for understanding and extracting key information from legal documents

- Designed and implemented end-to-end MLOps pipeline with CI/CD and monitoring
- Optimized retrieval via semantic search (vLLM + ChromaDB), achieving sub-200ms latency
- **Techstack:** GitHub Actions, FastAPI, vLLM, ChromaDB, Docker, Prometheus/Grafana

Publications

Towards Efficient Context-Aware Classification with Compact VLM Architectures: Indoor Fire Case Study

Apr 2026

Anh Tuan Giang, Nhat Quang Doan, **Minh Duong Tran**, et al.

<https://doi.org/10.1038/s41598-026-48743-5> [↗](#) (Scientific Reports [Q1 Journal])

Certifications

AWS Certified Solution Architect Associate (AWS-SAA-C03) [↗](#)